ChatGPT

Masterplan: AI Brain Workflow System (for Humans & AI Agents)

Overview: Architecture of the AI Brain

This document describes the core structure and functionality of the AI Brain System for the Heal the World Project. It provides a shared understanding for both human operators and AI agents.

Architecture Diagram (Conceptual)

```
[  Human/GUI]       [  Agent 1]        [  Agent 2]          [  ChatGPT Agent]
     |                   |                  |                       |
     | Trigger: DF/DFG   | Trigger: DF/DFG   | Trigger: DF/DFG       |
     |------------------>|------------------>|---------------------->|
     |                   |                  |                       |
     |   Reads:          |   Reads:          |   Reads:              |
     | -> masterplan.md  | -> state.json     | -> tasks/             |
     |                   |                  |                       |
     |   Executes work   | Executes task     | Monitors workflows    |
     |                   |                  |                       |
     |   Writes:         | Updates:          | Logs to:              |
     | -> logs/          | -> masterplan.md  | -> state.json         |
     |------------------> GitHub Repo <----------------------------->|
                       <--> PC local folder (sync)
```

🗂 Directory Structure (Mirrored: GitHub & Local PC)

```
/project-root/
├── masterplan/
│   └── masterplan.md        # Growing central knowledge brain
├── state/
│   └── state.json           # Workflow state and agent status
├── logs/
│   └── log-YYYY-MM-DD.md    # Daily logs of actions and triggers
├── agents/
│   └── agent1.py            # AI agent script
```

```
├── triggers/
│   └── DF_trigger.py        # Start trigger
│   └── DFG_trigger.py       # Stop trigger, update masterplan
├── config/
│   └── settings.yaml        # API keys, GitHub token, repo URL
└── gui/
    └── gui.py               # Human interface (GUI)
```

Trigger System (Workflow Control)

Trigger: DF (Start Workflow)

- Reads current `state.json`
- Identifies the next task
- Agents begin working on assigned actions
- Logs are created in `/logs`

Trigger: DFG (Stop & Summarize)

- Agents stop current task
- Write summaries to `masterplan/masterplan.md`
- Update `state.json` with new status
- Commit & push changes to GitHub

---

masterplan.md (Central Brain)

```
# 🌍 Heal the World Masterplan

##   2025-06-09
- Trigger: DF (by GUI)
- Agents: Agent1, Agent2
- Tasks: Drought analysis, logistics scan

###   Summary
Agent1 completed mapping of high-risk drought zones.
Agent2 ran simulation for aid delivery points.

##   Insights
- Use OSM overlays for faster mapping
- Prioritize unstable regions for logistics
```

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state.json (Workflow Status)

```
{
  "current_workflow": "drought-aid-africa",
  "step": "logistics-mapping",
  "agents_active": ["agent1", "agent2"],
  "last_trigger": "DF",
  "last_update": "2025-06-09T11:00Z"
}
```

Security & Sync

- GitHub Repo: Private, with token-based access (stored in `.env` or `settings.yaml`)
- Git Sync: On PC boot and on DFG trigger
- Auto Git script:

```
# Auto-sync script
cd ~/HTW-masterplan/
git pull origin main
git add .
git commit -m "Update from agent/GUI"
git push origin main
```

Agent Behavior Example

```
# agent1.py

def run():
    state = read_json("state/state.json")
    if state["current_workflow"] == "drought-aid-africa":
        result = analyze_drought_zones()
        write_log(result)
        update_masterplan(result)
        update_state("logistics-mapping", "done")
        push_to_github()
```

🖥️ GUI Interface (for Human Operator)

- Start DF Trigger
- Stop DFG Trigger
- Read masterplan
- Manual Git Sync
- 🗔 Open local directory
- (Optional): Text field to pass input to DF

This document will evolve as agents contribute to the `masterplan.md`, continuously growing the intelligence of the Heal the World system.